Out-of-time-order correlator (OTOC):

$$C_T(t) := -\langle [W(t), V(0)]^2 \rangle \quad , \langle \cdots \rangle : \text{thermal limit}$$

consider $x(t)$, $p(t)$:

$$C_T = -\langle [x(t), p(0)]^2 \rangle \xrightarrow{\text{semiclassical}} -\left\langle \left(i\hbar \underbrace{\frac{\delta x(t)}{\delta x(0)}}_{\sim e^{\lambda t}} \right)^2 \right\rangle$$

$$\Longrightarrow C_T \sim \hbar^2 e^{2\lambda t}$$

↳ quantum Lyapunov exponent

t_E : Ehrenfest time

A maximum bound of the quantum Lyapunov exponent λ was proposed as $\lambda \leq 2\pi k_B T/\hbar$. (Sachdev-Ye-Kitaev model)

OTOC in QM

H : time-independent Hamiltonian

$$C_T(t) := -\langle [x(t), p(0)]^2 \rangle \quad , \langle \mathcal{O} \rangle := \frac{\mathrm{tr}\,[e^{-\beta H}\mathcal{O}]}{\mathrm{tr}\, e^{-\beta H}} \quad , \beta = T^{-1}$$

Taking the energy eigenstates as basis :

$$C_T(t) = \frac{1}{Z} \sum_n e^{-\beta E_n} C_n(t) \quad , \quad C_n(t) = -\langle n | [x(t), p]^2 | n \rangle$$

↳ $p := p(0)$

OTOC for a fixed energy eigenstate : $C_n(t)$ microcanonical OTOC

Thermal OTOC = thermal average of microcanonical OTOCs

$$C_n(t) = -\langle n | [x(t), p][x(t), p] | n \rangle$$

$$= -\langle n | [x(t), p] | m \rangle \langle m | [x(t), p] | n \rangle$$

$$= \sum_m b_{nm}(t)\, b^*_{nm}(t) \quad , \quad b_{nm}(t) = -i \langle n | [x(t), p] | m \rangle$$

↳ Hermitian

$$b_{nm}(t) = -i \langle n | [e^{iHt} x e^{-iHt}, p] | m \rangle$$

$$= -i \langle n | e^{iHt} x e^{-iHt} p - p e^{iHt} x e^{-iHt} | m \rangle$$

$$= -i \sum_k \langle n | e^{iHt} x | k \rangle \langle k | e^{-iHt} p | m \rangle$$

$$+ i \sum_k \langle n | p e^{iHt} | k \rangle \langle k | x e^{-iHt} | m \rangle$$

$$= -i \sum_k e^{iE_n t - iE_k t} x_{nk} p_{km} + i \sum_k e^{iE_k t - iE_m t} p_{nk} x_{km}$$

$$= -i \sum_k \left(e^{iE_{nk} t} x_{nk} p_{km} - e^{iE_{km} t} p_{nk} x_{km} \right)$$

* For a natural $H = \sum_{i=1}^{N} p_i^2 + U(x_1, \dots, x_N)$

$[H, x] = -2ip \implies p_{mn} = \frac{i}{2} E_{mn} x_{mn}$

$\implies b_{mn}(t) = \frac{1}{2} \sum_k x_{nk} x_{km} \left(E_{km} e^{iE_{nk}t} - E_{nk} e^{iE_{km}t} \right)$

For actural numerical calculations, we need truncation for the $\sum$.

Integrable systems

① Harmonic oscillator:

$E_n = (n + \frac{1}{2})\omega, \quad x_{nm} = \frac{1}{\sqrt{\omega}} \left(\sqrt{m}\, \delta_{n,m-1} + \sqrt{m+1}\, \delta_{n,m+1} \right)$

$b_{nm}(t) = \delta_{nm} \cos \omega t$

$\implies C_n(t) = \cos^2 \omega t, \quad C_T(t) = \cos^2 \omega t$

② Particle in 1D box

$H = p^2 + V_{box}(x), \quad V_{box}(x) = \begin{cases} 0 & 0 < x < 1 \\ \infty & \text{else} \end{cases}$

$\psi_n = \sqrt{2} \sin \pi n x, \quad E_n = \pi^2 n^2, \quad n = 1, 2, \dots$

$x_{nm} = \begin{cases} \frac{1}{2} & n = m \\ \frac{1 - (-1)^{n+m}}{\pi^2} \left[\frac{1}{(n+m)^2} - \frac{1}{(n-m)^2} \right] & n \neq m \end{cases}$

Numerically, $n, m \leq N_{trunc} = 100$

③ Circle billiard

Take $R = \frac{1}{\sqrt{\pi}}$

$H = p_1^2 + p_2^2 + V_{circ}(x, y), \quad V_{circ}(x,y) = \begin{cases} 0 & x^2 + y^2 < R^2 \\ \infty & \text{else} \end{cases}$

$x = r\cos\theta, \quad y = r \sin\theta$

$\implies E_{kl} = \pi \rho_{kl}^2, \quad \psi_{kl} = \mathcal{N} J_k(\sqrt{\pi}\, \rho_{kl} r) e^{ik\theta}$

$k \in \mathbb{Z}, \quad l = \{1, 2, \dots\}$

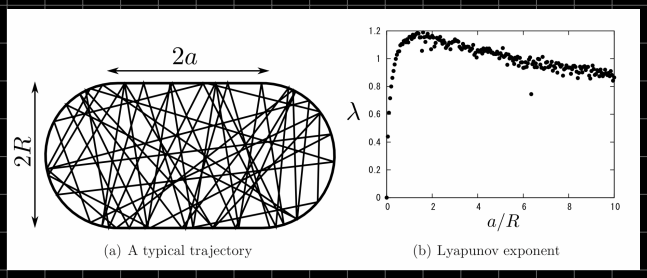
J_k: 1ST Bessel function, ρ_{kl}: its l-th root $\rightarrow J_k(\rho_{kl}) = 0$

$\mathcal{N} = \pi J_{k+1}(\rho_{kl})$

Non-integrable system : stadium billiards

① classical

$$H = p_1^2 + p_2^2 + V_{stad}(x, y), \qquad V_{stad}(x,y) = \begin{cases} 0 & (x,y) \in \Omega \\ \infty & \text{else} \end{cases}$$



(a) A typical trajectory (b) Lyapunov exponent

Inside : freely, constant velocity

Boundary : reflect elastically

phase space variable $X(t)$, linear perturbation $X(t) \to X(t) + \delta(t)$

$\delta(t) \sim e^{\lambda t}$ λ: Lyapunov exponent

* A positive Lyapunov exponent is the signal of chaos.

② QM of stadium billiards

$$[-\Delta + V_{stad}(x,y)]\psi_n = E_n \psi_n$$

$$E_n^{smooth} = \frac{4\pi}{A} n + \frac{(4\pi)^{\frac{1}{2}} \partial A}{A^{\frac{3}{2}}} \sqrt{n}$$, ∂A : length of boundary

$$C(r,t) = \langle [W_0(t), V_r]^\dagger [W_0(t), V_r] \rangle$$

$$W_0(t) = e^{iH_0 t} W_0 e^{-iH_0 t}$$

OTOC : $F(t) = \langle W_0^\dagger(t) V_r^\dagger W_0(t) V_r \rangle$ $\qquad$ W, V : unitary

$$C = 2(1 - \mathrm{Re}\, F)$$

$$|\Phi\rangle = \prod_{r=1}^{n} \frac{|00\rangle_{r\tilde{r}} + |11\rangle_{r\tilde{r}}}{\sqrt{2}}$$

$|W\rangle$ is obtained by $|W\rangle = W \otimes I \; |\Phi\rangle$

Lieb-Robinson $\longrightarrow$ light cone